The Curve Estimates for Proper Base Change a brief blueprint

(Blueprint 06, brief version — every node of the handwritten plan is its own proposition: Tsen's theor

Abstract

This is the brief blueprint of the *curve-estimate* step behind proper base change (the statement marked "split marker" in the PBC blueprint; `OBL-PBC-CURVES` in the project). Per the readme conventions, *every node of the handwritten plan is a separate proposition* (no bundled conclusions): Tsen's theorem and the divisor sequence are stated on their own, they produce $\mathrm{Br}(C) = 0$, the Kummer sequence gives the two identifications $H^1(C, \mu_n) = \mathrm{Pic}(C)[n]$ and $H^2(C, \mu_n) = \mathrm{Pic}(C)/n\mathrm{Pic}(C)$ as two statements, the **Jacobian** (input: $\mathrm{Pic}/\mathrm{Pic}^0$ representable, as requested) and the **abelian-variety** facts are split off, and the target theorem is last. Statements list definitions in `\uses`, proofs cite with `\Cref`, and steps whose split may be very large carry qualitative **split markers**.

1 Definitions

Definition 1.1 (Curves and genus)**.** Let k be algebraically closed. A *curve* over k is a separated scheme of finite type over k of dimension ≤ 1; it is *smooth* (resp. *proper*) if its structure morphism is smooth (resp. proper). For C smooth proper connected, the *genus* of C is $g = \dim_k H^0(C, \Omega^1_{C/k})$ (invariant under algebraically closed extension of k).

Definition 1.2 (The Kummer sequence)**.** *Uses:* `def:curve`

For $n \geq 1$ and C as in 1.1 Curves and genus, the *Kummer sequence* is the sequence of étale sheaves on C

$$0 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{\omega \mapsto \omega^n} \mathbb{G}_m \longrightarrow 0$$

(an n-th root of a unit exists étale locally).

Definition 1.3 (Picard group, degree, Pic^0)**.** *Uses:* `def:curve`

For C as in 1.1 Curves and genus: $\mathrm{Pic}(C) = H^1(C_{\text{ét}}, \mathbb{G}_m)$ (invertible $\mathcal{O}_C$-modules up to isomorphism). For C smooth proper connected there is a *degree* homomorphism $\deg : \mathrm{Pic}(C) \to \mathbb{Z}$ and we set $\mathrm{Pic}^0(C) = \ker \deg$; for a group f and $n \geq 1$, $f[n]$, $n \cdot f$, $f/n \cdot f$ denote kernel, image and cokernel of the multiplication-by-n map.

Definition 1.4 (Brauer group)**.** *Uses:* `def:curve`

For C as in 1.1 Curves and genus, the *Brauer group* of C is $\mathrm{Br}(C) = H^2(C_{\text{ét}}, \mathbb{G}_m)$.

2 The propositions

2.1 Tsen's theorem

Proposition 2.1 (Tsen's theorem)**.** *Uses:* `def:curve, def:br`

For C smooth over algebraically closed k, the function field $k(C)$ is a C_1 field. Tsen's theorem and its standard Galois-cohomological consequence give

$$H^r(k(C), \mathbb{G}_m) = 0 \quad (r \geq 1), \qquad \mathrm{Br}(k(C)) = H^2(k(C), \mathbb{G}_m) = 0.$$

SPLIT MARKER — Tsen's theorem ($k(C)$ is C_1; Tsen 1933; Stacks 03RD, 03RF) may itself split into a very large number of statements (C_1-stability under finite extensions, norm forms, division algebras).

Idea. *Uses:* `def:br`

Tsen says that $k(C)$ is C_1. Hilbert 90 and the standard Galois-cohomological consequence of the C_1 property give $H^r(k(C), \mathbb{G}_m) = 0$ for $r \geq 1$ (in particular, $\mathrm{Br}(k(C)) = 0$ by 1.4 Brauer group). □

2.2 The divisor sequence

Proposition 2.2 (Divisor sequence and the generic-fibre injection)**.** *Uses:* *def:curve, def:br*

For C as in 1.1 Curves and genus with generic point η and inclusion $j : \eta \hookrightarrow C$, there is an exact sequence of étale sheaves on C

$$0 \longrightarrow \mathbb{G}_m \longrightarrow j_*\mathbb{G}_{m,\eta} \longrightarrow \bigoplus_{p \in C} i_{p*}\mathbb{Z} \longrightarrow 0,$$

and consequently (Grothendieck's residue sequence) the base change map is injective:

$$\mathrm{Br}(C) = H^2(C, \mathbb{G}_m) \hookrightarrow H^2(\eta, \mathbb{G}_m) = \mathrm{Br}(k(C)).$$

SPLIT MARKER — the divisor exact sequence and the residue sequence for $H^2(X, \mathbb{G}_m)$ (Dix Exposés) may split into a very large number of statements.

Idea. The sequence is exact by the structure of meromorphic functions on a curve (poles with multiplicities); its long exact sequence, with H^0 of the middle and third terms being the divisor group, gives $H^0(G_m, C) = k^\times$, H^1 the Picard group and the asserted injection in degree 2 (Grothendieck's residue sequence). □

2.3 Brauer group of a curve

Theorem 2.3 ($\mathrm{Br}(C) = 0$)**.** *Uses:* *def:curve, def:br, prop:tsen, prop:divisor*

For C smooth proper over algebraically closed k: $\mathrm{Br}(C) = 0$.

Proof. *Uses:* `prop:tsen`, `prop:divisor`

$\mathrm{Br}(C) \hookrightarrow \mathrm{Br}(k(C)) = 0$ by 2.2 Divisor sequence and the generic-fibre injection and 2.1 Tsen's theorem. □

2.4 The two Kummer identifications

Proposition 2.4 (H^1 via Kummer)**.** *Uses:* *def:kummer, def:pic, def:br, thm:br*

For C as in 1.1 Curves and genus and every $n \geq 1$:

$$H^1(C, \mu_n) \cong \mathrm{Pic}(C)[n].$$

Idea. *Uses:* `thm:br`, `def:kummer`, `def:pic`

The Kummer long exact sequence of 1.2 The Kummer sequence reads

$$0 \to \mu_n(k) \to k^\times \xrightarrow{n} k^\times \to H^1(C, \mu_n) \to \mathrm{Pic}(C) \xrightarrow{n} \mathrm{Pic}(C) \to H^2(C, \mu_n) \to H^2(C, \mathbb{G}_m) = \mathrm{Br}(C),$$

where $k^\times \to k^\times$, $x \mapsto x^n$, is surjective (k algebraically closed) and $H^1(C, \mathbb{G}_m) = \mathrm{Pic}(C)$ by 1.3 Picard group, degree, Pic^0. The last term is 0 by 2.3 $\mathrm{Br}(C) = 0$, so exactness gives $H^1(C, \mu_n) = \ker(n : \mathrm{Pic}(C) \to \mathrm{Pic}(C)) = \mathrm{Pic}(C)[n]$. □

Proposition 2.5 (H^2 via Kummer)**.** *Uses:* *def:kummer, def:pic, def:br, thm:br*

For C as in 1.1 Curves and genus and every $n \geq 1$:

$$H^2(C, \mu_n) \cong \mathrm{Pic}(C)/n \cdot \mathrm{Pic}(C).$$

Proof. *Uses:* `thm:br`, `def:kummer`, `def:pic`

In the same long exact sequence of 1.2 The Kummer sequence (with $H^1(C, \mathbb{G}_m) = \mathrm{Pic}(C)$ by 1.3 Picard group, degree, Pic^0 and the last term 0 by 2.3 $\mathrm{Br}(C) = 0$) the displayed quotient is the cokernel of $n : \mathrm{Pic}(C) \to \mathrm{Pic}(C)$. □

2.5 The Jacobian (extra input: $\mathrm{Pic}/\mathrm{Pic}^0$ representable)

Proposition 2.6 (Jacobian: Pic^0 is representable — input)**.** *Uses:* def:curve, def:pic

For C smooth proper connected of genus g over algebraically closed k the functor Pic^0 is representable: there is an abelian variety $J(C)$ of dimension g — the Jacobian *— with a canonical identification $\mathrm{Pic}^0(C) \cong J(C)(k)$, compatible with algebraically closed base extension.*

SPLIT MARKER — deep input, quoted (Stacks 0B92/0B9Z, Mumford/Milne, Abelian Varieties): Abel–Jacobi, rigidity, the Picard scheme of a curve. This is the requested extra input: representability of $\mathrm{Pic}/\mathrm{Pic}^0$.

Omitted — quoted input. The representability theorem is quoted; in the blueprint it is the Jacobian theorem of `06_smooth_curve_calculations.tex` (`thm:sc-jacobian-representability`). □

2.6 Abelian varieties (split off)

Proposition 2.7 (The multiplication-by-n isogeny)**.** *Uses:* def:curve

Let A be an abelian variety of dimension g over algebraically closed k and let $n \geq 1$. The multiplication map $[n] : A \to A$ is an isogeny: finite, surjective, of degree n^{2g}.

Idea. $[n]$ is a surjective endomorphism of finite degree n^{2g} (divisibility of a connected commutative group scheme; the degree formula $\deg[n] = n^{2g}$); finite and surjective by the theory of isogenies. □

Proposition 2.8 (Étale n-torsion)**.** *Uses:* prop:av-isogeny

Let A be an abelian variety of dimension g over algebraically closed k and let $n \geq 1$ be prime to $\mathrm{char}\, k$*. Then $A[n]$ is a finite étale group scheme of rank n^{2g}, so*

$$A[n] \cong (\mathbb{Z}/n)^{2g}.$$

SPLIT MARKER — the abelian-variety block (rigidity, commutativity, the theorem of the cube, $[n]^*L$, $\deg[n] = n^{2g}$, étale n-torsion, the p-power fibre bound, Cartier's theorem) may split into a very large number of statements.

Proof. *Uses:* prop:av-isogeny

$A[n] = \ker[n]$ has rank $\deg[n] = n^{2g}$ by 2.7 The multiplication-by-n isogeny; it is étale because n is prime to the characteristic (the n-torsion is étale by the theory of the n-isogeny); a finite étale group scheme over algebraically closed k is $(\mathbb{Z}/n)^{2g}$ (it is killed by n and has rank n^{2g}). □

Proposition 2.9 (Divisibility)**.** *Uses:* prop:av-isogeny

With the hypotheses of 2.7 The multiplication-by-n isogeny, $A/n \cdot A = 0$.

Proof. *Uses:* prop:av-isogeny

By 2.7 The multiplication-by-n isogeny, $[n] : A \to A$ is surjective, so every element of A is n times an element of A. □

2.7 Cohomological dimension

Proposition 2.10 (Vanishing in high degrees)**.** *Uses:* def:curve, def:kummer, prop:divisor, prop:tsen

For C a smooth curve over algebraically closed k and n prime to $\mathrm{char}\, k$*:*

$$H^q(C, \mathbb{Z}/n) = 0 \qquad (q \geq 3).$$

Proof. *Uses:* `def:kummer`, `prop:divisor`, `prop:tsen`

Put $K = k(C)$ and let $j : \eta = \operatorname{Spec} K \hookrightarrow C$ be the generic point. By 2.2 Divisor sequence and the generic-fibre injection,

$$0 \longrightarrow \mathbb{G}_m \longrightarrow j_*\mathbb{G}_{m,K} \longrightarrow \bigoplus_{x \in C_{\mathrm{cl}}} i_{x*}\mathbb{Z} \longrightarrow 0.$$

By 2.1 Tsen's theorem, $H^r(\eta, \mathbb{G}_m) = 0$ for $r \geq 1$; at a closed geometric point the stalks of $R^r j_*\mathbb{G}_{m,K}$ are the same groups over an algebraic extension of K, and the generic stalk is over a separably closed field. Hence $R^r j_*\mathbb{G}_{m,K} = 0$ for $r \geq 1$. Leray therefore gives $H^r(C, j_*\mathbb{G}_{m,K}) = 0$ for $r \geq 1$. Each i_x is finite and $x = \operatorname{Spec} k$, so

$$H^r\left(C, \bigoplus_x i_{x*}\mathbb{Z}\right) = 0 \qquad (r \geq 1).$$

The long exact sequence thus gives

$$H^r(C, \mathbb{G}_m) = 0 \qquad (r \geq 2).$$

Finally, the Kummer sequence of 1.2 The Kummer sequence gives, for $q \geq 3$,

$$H^{q-1}(C, \mathbb{G}_m) \longrightarrow H^q(C, \mu_n) \longrightarrow H^q(C, \mathbb{G}_m).$$

Both outer groups vanish, so $H^q(C, \mu_n) = 0$. Choosing a primitive n-th root of unity identifies $\mu_n \cong \mathbb{Z}/n$, proving the claim. □

3 The final theorem (the curve estimate)

Theorem 3.1 (Curve estimate)**.** *Uses:* *`def:curve`, `def:pic`, `def:kummer`, `prop:h1`, `prop:h2`, `prop:jacobian`, `prop:av-ntorsion`, `prop:av-divisible`, `prop:cd`*

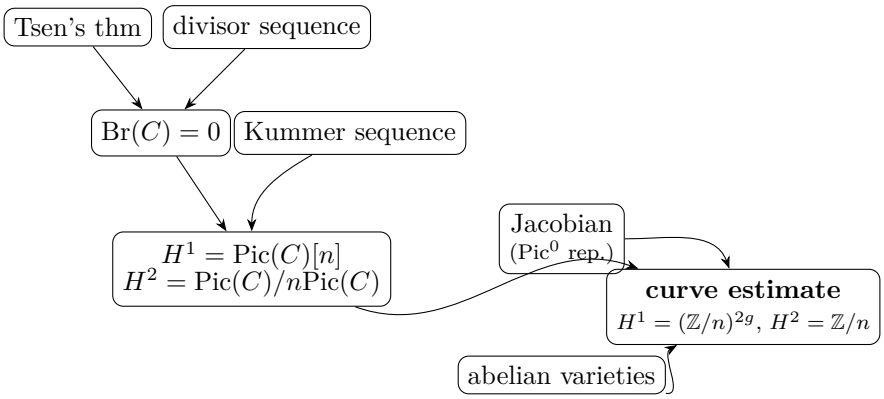
Let C be smooth proper connected of genus g over algebraically closed k, and let n be prime to $\operatorname{char} k$. After choosing a primitive n-th root of unity (the identification $\mu_n \cong \mathbb{Z}/n$ of étale sheaves), the constant-sheaf cohomology of C is

$$H^0(C, \mathbb{Z}/n) = \mathbb{Z}/n, \qquad H^1(C, \mathbb{Z}/n) = (\mathbb{Z}/n)^{2g}, \qquad H^2(C, \mathbb{Z}/n) = \mathbb{Z}/n,$$

and $H^q(C, \mathbb{Z}/n) = 0$ for $q \geq 3$.

Proof. *Uses:* `prop:h1`, `prop:h2`, `prop:jacobian`, `prop:av-ntorsion`, `prop:av-divisible`, `prop:cd`

$H^0(C, \mathbb{Z}/n) = H^0(C, \mu_n) = \mu_n(k) = \mathbb{Z}/n$ (algebraically closed, primitive root). By 2.4 H^1 via Kummer, $H^1(C, \mathbb{Z}/n) = \operatorname{Pic}(C)[n]$; the degree map splits $\operatorname{Pic}(C) = \operatorname{Pic}^0(C) \times \mathbb{Z}$ (a base point gives a section) and $\mathbb{Z}$ is torsion-free, so $\operatorname{Pic}(C)[n] = \operatorname{Pic}^0(C)[n] = J(C)[n]$, which is $(\mathbb{Z}/n)^{2g}$ by 2.6 Jacobian: Pic^0 is representable — input and 2.8 Étale n-torsion. By 2.5 H^2 via Kummer, $H^2(C, \mathbb{Z}/n) = \operatorname{Pic}(C)/n\operatorname{Pic}(C) = (\mathbb{Z}/n) \oplus J(C)/nJ(C) = \mathbb{Z}/n$ by 2.9 Divisibility (the $\mathbb{Z}/n$ is the degree part). The vanishing is 2.10 Vanishing in high degrees. □



Dependency graph (arrows = `\Cref`-dependencies): 2.1 Tsen's theorem and 2.2 Divisor sequence and the generic-fibre injection give $\mathrm{Br}(C) = 0$ (2.3 $\mathrm{Br}(C) = 0$), which together with the Kummer sequence of 1.2 The Kummer sequence yields the two identifications 2.4 H^1 via Kummer and 2.5 H^2 via Kummer; 2.6 Jacobian: Pic^0 is representable — input (input: $\mathrm{Pic}/\mathrm{Pic}^0$ representable) and 2.8 Étale n-torsion/2.9 Divisibility turn these into $(\mathbb{Z}/n)^{2g}$ and $\mathbb{Z}/n$ of 3.1 Curve estimate; 2.10 Vanishing in high degrees gives the vanishing.

Remark 3.2. *Where this feeds into proper base change.* The estimate of 3.1 Curve estimate is the constant-sheaf instance of the calculations behind the PBC curve-estimate statement (theorem 2.4 of the PBC blueprint): via $\mu_n \cong \mathbb{Z}/n$, trace/duality and 2.10 Vanishing in high degrees it gives the assertion block (C_1)–$(C_6)/(C_8)$ for constant sheaves; the split continues (normalization, constructible and dense-open reduction, locally constant reduction, dévissage) to all torsion sheaves, thence to the invariance under separably closed extension and to $\mathrm{PBC}(\mathbf{P}^1)$. *Every node is a separate statement*, per the handwritten plan and the readme's separation-of-conclusions rule; the steps that may split into very many statements are marked qualitatively: Tsen (2.1 Tsen's theorem), the divisor/residue sequences (2.2 Divisor sequence and the generic-fibre injection), the Jacobian representability input (2.6 Jacobian: Pic^0 is representable — input), the abelian-variety block (2.8 Étale n-torsion); the cohomological-dimension proposition (2.10 Vanishing in high degrees) is now proved explicitly and carries no split marker. Corresponding blueprint source: `06_smooth_curve_calculations.tex` (target `OBL-PBC-CURVES`).